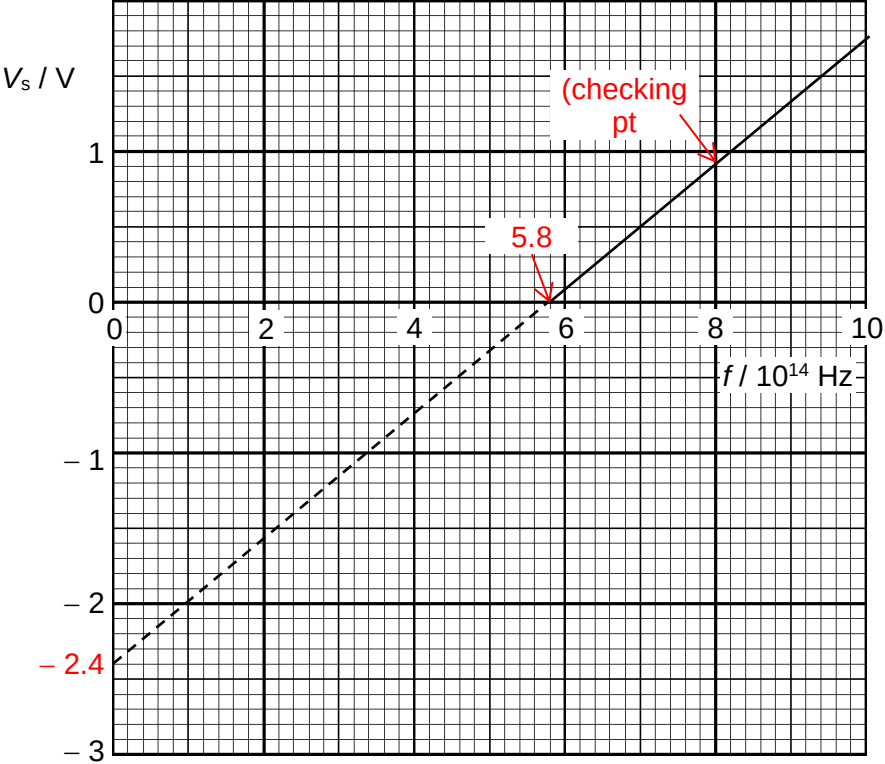
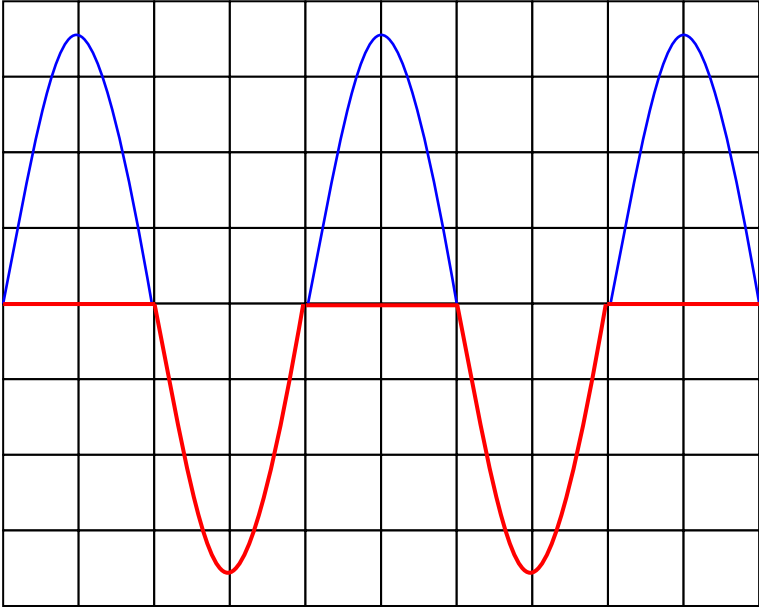


Qns	Answer	Marks
5(a)	$hf = \Phi + \frac{1}{2}mv_{\max}^2$ $= \Phi + eV_s$ $V_s = \left(\frac{h}{e}\right)f - \frac{\Phi}{e}$ when $V_s = 0$, $f = \frac{\Phi}{h}$ $= \frac{2.4(1.6 \times 10^{-19})}{6.63 \times 10^{-34}} = 5.79 \times 10^{14} \text{ Hz}$ when $f = 0$, $V_s = -\frac{\Phi}{e}$ $= -\frac{2.4 \text{ eV}}{e} = -2.4 \text{ V (extrapolated)}$ 	1 cut x-axis at (5.8, 0) 1 dotted line to (0, -2.4) OR pass through (8, 0.9)
5(b)	$V_{\text{peak, supply}} = \sqrt{2} V_{\text{supply, rms}}$ $= \sqrt{2}(4.6) = 6.5 \text{ V}$ $V_{\text{peak, diode}} = V_{\text{peak, supply}}$ $= 6.5 \text{ V}$	1

Qns	Answer	Marks
5(c)		1

Qns	Answer	Marks
c(Vi)	time spent between plates	